AMAZON BASE BUILDING COMPLIANCE STANDARD

Task Title	Photovoltaic Installation
APM Object	CT.ELEC.PHOTOV
APM Description	Photovoltaic installations
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure. CT evidence must be additionally uploaded to the DMS (WorkDocs in DE) To suggest improvements please raise a SIM or contact document authors Leonardo Vitale (LEOVITAL) Rob Gisby (EWARTGXR) or APM, Benito Sanchez Sanchez de Puerta (BPPUERTA) SIM - Link to SIM template

Last Update: (Version)	ago-23	**Last Update Change Detail:**	Transfer to new template. Rationalisation of EU RME Minimum standards. Alignment with dependant PM. Released as Draft.	**Activities may be covered by No-Go restrictions:**	Yes	**Special Certification Required:**	**Delivered By:**
Evidenced part or full by PM:	Not Specified	**Related BBM PM:**	N/A	**Related CT Areas:**	Not Specified		

Scope

Any site where photovoltaic installations are used to generate solar power.
The standard covers only the components of the installation, the electrical elements are covered in CT.ELEC.ELECINST and CT.ELEC.FIXEDWIRE

Using this document

This document is intended as a guide and may not be exhaustive. Whilst this document is based extensively on professional guidance and subject to periodic review, individual sites must be diligent to ensure they are aware of, and comply to, local and changing regulations.
The format of these standards is intended to improve navigation between maintenance and compliance standards and to allow frequent, minor updates to a single living resource. Please avoid offline or printed versions and check version dates and change log.
In the event of becoming aware of errors or omissions in this document, sites must provide detail, including references to relevant source information to their local planner for escalation - SIM Link above.
This document is considered released as draft.
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Maintenance and Compliance Relationship

Where relevant, associated Base Building PM's are listed on the header . In many cases maintenance contributes to, or fulfils compliance requirement. This may not be exhaustive and is intended to assist users to identify relevant maintenance.
If a BBM Maintenance activity fully meets the compliance requirements below, periodic upload of PM evidence will be required to APM and DMS / Workdays depending on region.

Key to Abbreviations

The key to the below document is as follows:
CT - Amazon compliance / statutory task - not otherwise covered by a BBM Standard PM. Evidence to be provided into APM and DMS upon task closure.
CT* - This is a conditional task based or based on findings from previous inspections. If a frequency is not used, a further appropriate frequency needs to be in either the region, or 'ALL'. Local teams must ensure appropriate frequency.
(CT) - Compliance task which is likely met or exceeded by a CT task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met.
PM - Requirement can demonstrated through appropriate maintenance - Refer to relevant PM Standard
(PM) - Maintenance task which is met or exceeded by a PM task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met including regional / manufacturer requirements.
EVID - Periodic upload of evidence to DMS, for non-CT task lines. For example, evidence of PM completion, certification, licence etc.
EVID* - Conditional requirement to upload evidence at the interval provided
N A R - No Additional Requirement

Required Tasks

Region	Task Index	Task Frequency	Task Requirements	Qualification of Inspector	Records Required	PM	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	Additional
All	1	In accordance with manufacturer's recommendations and instructions, and with consideration of site risk assessment but at least annually. Interim inspections will be required following adverse	Regular inspection of condition of components (PV cells including mounts, batteries, wiring, balance of system equipment) for cracks, chips, discolouration and corrosion (e.g. to mounting frame or battery terminals). Schedule resultant remedial actions as required.	Competent person.	No additional requirements						CT		
All	2	In accordance with manufacturer's recommendations and/or where efficiency has been observed to fall.	Cleaning of PV Cells (if required - N.B some installations are designed to be self cleansing).	Competent person.	No additional requirements						CT*		
All	3	Calibration of photovoltaic meters .	In accordance with manufacturer's recommendations and instructions.	Competent person.	No additional requirements						CT*		
DE	4	Every 16 years	Calibration of mechanical meters with induction movement	Competent person. State approved inspection body.	No additional requirements								16 Year CT
DE	5	Every 12 years	Calibration of mechanical transducer meters with induction movement, rotor disc	Competent person. State approved inspection body.	No additional requirements								12 Year CT

DE	6	Every 8 years	Calibration of electronic meters	Competent person. State approved inspection body.	No additional requirements								8 Year CT
FR	7	Every year and after climatic event.	Annual inspection of the equipment and safety elements of the photovoltaic production unit.	Competent person	No additional requirements						CT		

Related Regional Legislation			
Region	**Relevant legislation**	**Vendor Certification**	**Statutory requirement**
CZ	From acon No. 505/1990 Coll.	Certified laboratory	N/A
DE	DIN VDE 0105-100 - Betrieb von elektrischen Anlagen DIN VDE 0100-600 VDE 0100-600:2017-06 - Errichten von Niederspannungsanlagen Teil 6: Prüfungen RBS 1201 - Prüfungen und Kontrollen von Arbeitsmitteln und üb DGUV Vorschrift 3 - Unfallverhütungsvorschrift Elektrische Anlagen und Betriebsmittel DIN EN 62446-1 VDE 0126-23-1:2019-04 - Photovoltaik (PV)-Systeme – Anforderungen an Prüfung, Dokumentation und Instandhaltung	Testing organisation or specialist electrical company (Prüforganisation oder einen Elektrofachbetrieb)	Comprehensive inspection at least every 4 years, as well as additional inspections based on insurance requirements and risk assessment. Recommended: Daily visual inspection of the display to check the functionality of the inverter system by the operator. Daily check of remote monitoring system by the operator. Monthly check by operator to check and document the meter reading.
ES	RD 842/2002, of August 2, which approves the electrotechnical regulation for low voltage and its 52 Complementary Technical Instructions. ITC-BT-40. Low voltage generator facilities RD244/2019, of April 5, which establishes regulation about technical, economic, and administrative conditions for the for self-supply of electric energy. RD413/2014, of June 6, which establishes regulation about energy produced from renewable energy, cogeneration, and waste.	Accredited company (OCA)	Low voltage Inspections and testing at least every 5 years.
FR	Ministerial Order of October 4th, 2010 Article 16 to 23 https://www.legifrance.gouv.fr/loda/id/JORFTEXT000023081900/ The provisions shall apply to solar photovoltaic electricity generation equipment on the roof, façade or ground of a classified installation subject to authorisation. Solar photovoltaic electricity generation equipment is not subject to the requirements of this regulation if an analysis shows that it has no significant impact on the classified installation.	Competent Person	Every year and after climatic event.
IT	Legislative Decree 8/04/2008 n.81 DM 22/01/2008 n.37 CEI EN IEC 62446-2 (Maintenance of photovoltaic installations) CEI 0-16 (Interface systems-high and medium voltage) CEI 0-21 (Interface systems-low voltage) Resolution AEEG 786/2016/R/eel	Competent, accredited person	Meter's inspection every 5 years (electrodynamics meters) or 3 years (static meters). Interface systems maintenance every 5 years for installations over 11,08 kW. General cleaning and maintenance in accordance with manufacturer's instructions, best practice is every 12 months.
PL	Act of 7 July 1994 - Construction Law (Journal of Laws 1994 No. 89 pos. 414), as amended Act of April 10, 1997 - Energy Law (Journal of Laws of 1997, No. 54, item 348) Regulation of the Minister of Economy, Labor and Social Policy of 28 April 2003 on the detailed rules for confirming qualifications by persons involved in the operation of devices, installations and networks (Journal of Laws of 2003, No. 89, item 828) Regulation of the Minister of Energy of August 28, 2019 on health and safety at work with energy devices (Journal of Laws 2019, item 1830)	N/A	Inspections and testing at least every 5 years.
ROI	Safety, Health and Welfare at Work Act 2005	Competent Person	Provision and maintenance of plant and systems of work that are, so far as is reasonably practicable, safe and without risks to health.
SK	Act No. 124/2006 Coll. on safety and health at work Ordinance No. 508/2009 Coll. on details for ensuring safety and health protection at work with pressure, lifting, electrical and gas technical equipment	Qualified and certified person	Twice per year in case of temporary installations. Other installations with lower frequency.

UK	Health and Safety at Work etc. Act 1974 The Health and Safety at Work (Northern Ireland) Order 1978	Competent Person	Provision and maintenance of plant and systems of work that are, so far as is reasonably practicable, safe and without risks to health.